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SOFTWARE ENGINEERING MINI PROJECT- REPORT on HOSTEL ROOM ALLOCATION SYSTEM

Submitted in partial fulfillment of the requirements for the VI semester

Bachelor of Engineering

in

INFORMATION SCIENCE AND ENGINEERING

Submitted by

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2025-2026

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PROBLEM STATEMENT

The manual management of hostel operations involves complex, paper-based record-keeping that is inherently inefficient. Traditional systems are prone to several critical issues:

- **Data Redundancy:** Multiple entries for the same student or room are often scattered across different physical registers, leading to data inconsistency.
- **Inaccuracy:** Manual calculations are susceptible to errors when determining room vacancy or tracking visitor exit times.
- **Slow Processing:** The room allocation process is time-consuming and lacks the transparency required for students to track their application status.
- **Lack of Security:** Physical registers are easily accessible to unauthorized individuals and are difficult to back up or recover in case of damage.

ABSTRACT

The **Hostel Room Allocation System** is a web-based management solution designed to automate administrative tasks and streamline hostel operations. Developed using the **Node.js** ecosystem and **SQLite**, it provides a centralized platform for administrators to manage infrastructure, approve student applications, and monitor visitor activity. The system allows students to register online and track their allocation status, ensuring a transparent and efficient digital workflow.

OBJECTIVES

- **Automation:** To replace manual room allocation with a logic-driven, automated approval process.
- **Data Integrity:** To utilize a relational database to ensure consistent, accurate, and non-redundant records for hostels, rooms, and students.
- **Transparency:** To provide students with a dedicated dashboard to view registration status and allocated room details in real-time.

STAKEHOLDERS/ USERS/ INVESTORS

- **Administrator (Hostel Warden/Manager):** Responsible for managing room inventory, approving or rejecting student registrations, and monitoring security logs.
- **Students:** Primary users who apply for accommodation, manage their personal profiles, and check allocation status via a secure login.
- **Security Personnel:** Secondary users responsible for recording real-time visitor entries and exits through the system's dedicated visitor module.

TEAM MEMBERS AND ROLES

#	Name	Role	Responsibility
1	G Bharath	Product Owner	Defines project requirements and overall system objectives
2	Gahan Krishna K	Team Lead	Manages team coordination and ensures timely completion
3	Amith K N	Developer	Develops backend API logic and database integration
4	Hemanth K S	Designer	Designs the UI and improves user experience
5	Chirag H S	Tester	Tests modules and identifies bugs or errors
6	Basvaraj	Scrum Master	Coordinates Agile sprints and team activities

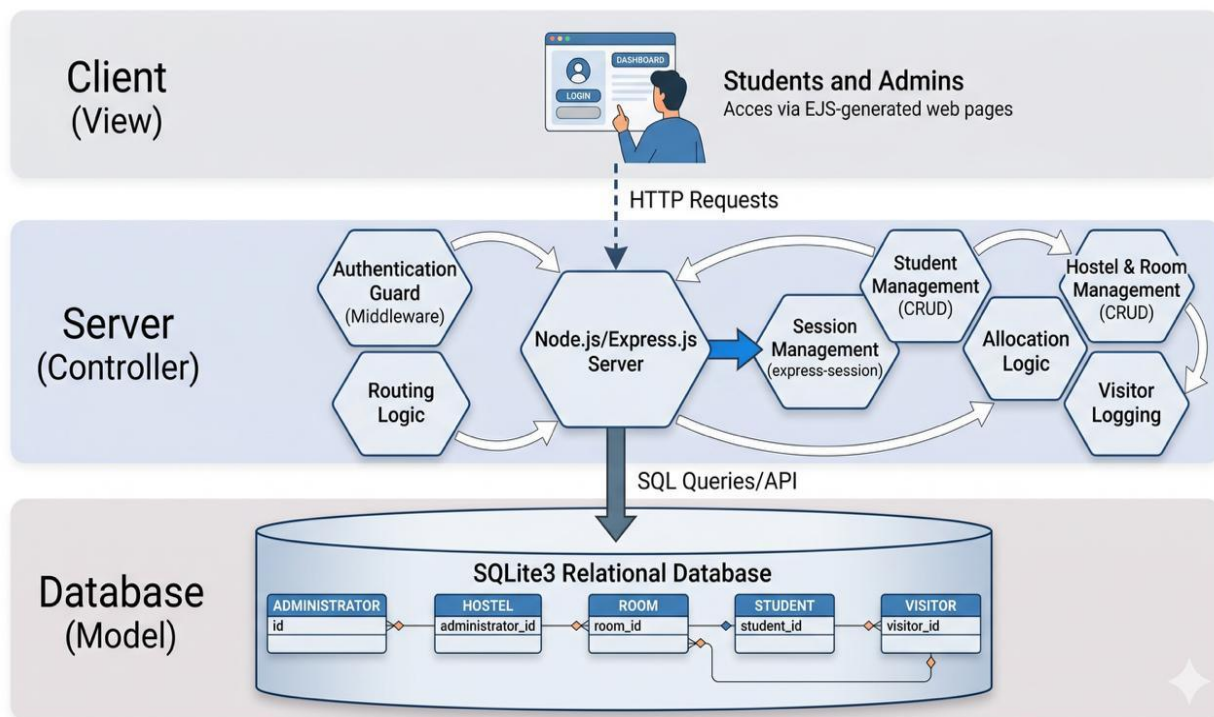
SYSTEM OVERVIEW

The system is a multi-user web application that digitizes the entire hostel lifecycle through the following workflow:

1. **Registration:** Students submit digital applications containing academic and contact details.

2. **Approval Workflow:** Administrators review "Pending" applications and approve them based on real-time room capacity.
3. **Allocation:** Upon approval, the system automatically identifies vacant rooms and assigns them to the student.
4. **Visitor Management:** A dedicated module tracks outsiders, maintaining timestamped logs of their visit duration.

SYSTEM ARCHITECTURE DIAGRAM



Frontend (Client)	Backend (Server)	Database (MySQL)
HTML5 ,CSS3,EJS Templatates	Node.js + Express.js	SQLite

TECHNOLOGY STACK

- **Backend:** Node.js, Express.js
- **Frontend:** HTML5, CSS3, EJS Templates
- **Database:** SQLite (Relational DBMS)
- **Session Management:** express-session for secure user authentication.
- **Environment:** NPM (Node Package Manager) for dependency and package management.

REQUIREMENTS

Functional Requirements

- Secure Login/Authentication for Admins and Students.
- Full CRUD (Create, Read, Update, Delete) operations for Hostels and Rooms.
- Student registration and status tracking (Pending/Approved/Rejected).
- Automated calculation of room occupancy and vacancy levels.

Non-Functional Requirements

- **Security:** Implementation of session-based authentication to protect admin routes.
- **Reliability:** Use of database constraints (Foreign Keys) to maintain data consistency.
- **Usability:** A minimalist and intuitive UI designed for ease of use by non-technical staff.

USE CASE MODELLING

- **Admin Use Cases:** Add/Delete Hostels, Modify Room Capacity, Approve Applications, View Student List, Log Visitors.
- **Student Use Cases:** Self-Registration, Login, View Dashboard, Check Room Number.
- **System Logic:** When an Admin approves a student, the system automatically queries the ROOM table for entries where `current_occupancy < capacity` and updates the student's status accordingly.

SYSTEM DESIGN

Component	Details & Specifications
User Roles	Students (Register/View Status), Admins (Approve/Manage Rooms), and Security (Visitor Logging).
Presentation Layer	Developed using EJS (Embedded JavaScript) for dynamic content and CSS3 for a minimalist, user-friendly UI.
Application Layer	Node.js & Express.js logic that manages secure sessions, protects admin routes, and handles the room allocation algorithm.
Data Layer	A relational SQLite3 database with five core tables: ADMINISTRATOR, HOSTEL, ROOM, STUDENT, and VISITOR.
Core System Logic	Upon Admin approval, the system verifies <code>current_occupancy < capacity</code> before assigning a room and updating the student status.

AGILE METHODOLOGY

The system is developed using the Agile methodology — an iterative and flexible approach focused on continuous improvement, teamwork, and delivering working software in small increments. Development was divided into sprints, each targeting specific features.

Agile Process Followed

The project was developed using the **Agile-Scrum** approach across four distinct sprints:

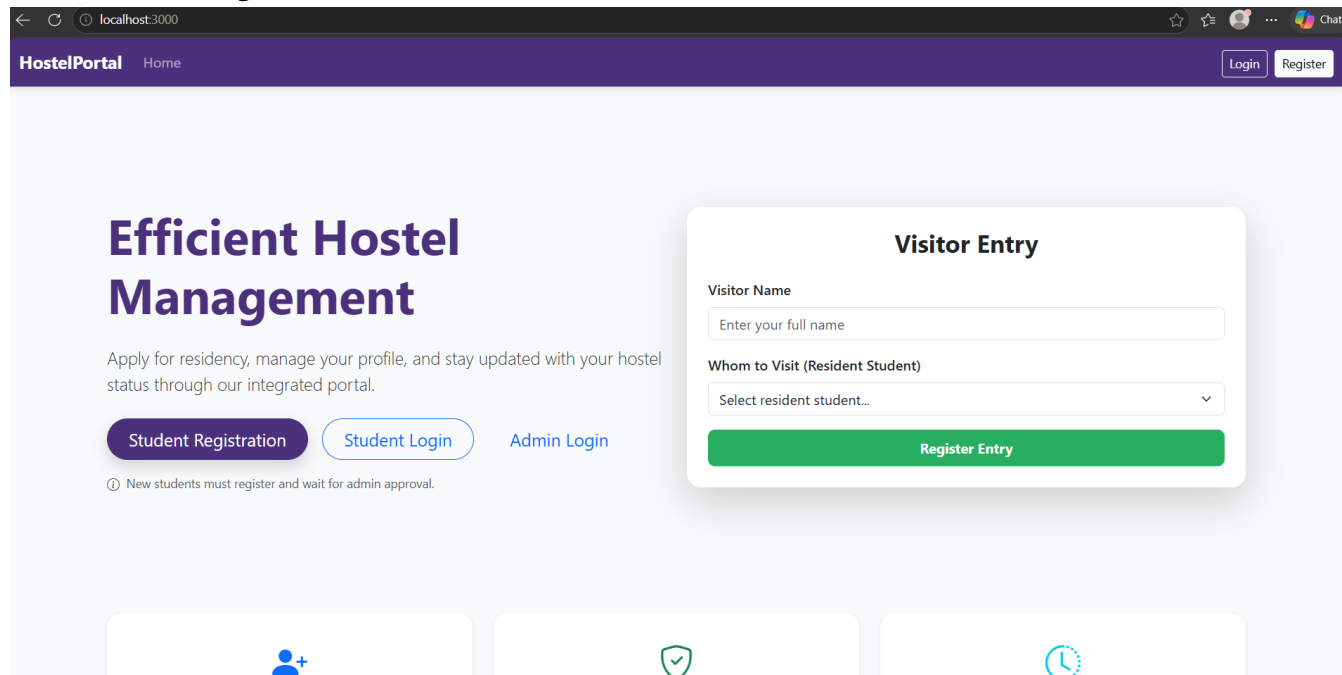
- **Sprint 1 (Requirement Gathering):** Identification of database entities (Student, Room, Hostel) and core feature sets.
- **Sprint 2 (Database Design):** Establishing the SQLite schema and seeding the database with initial test data.
- **Sprint 3 (Development):** Building Express routes and EJS views for core administrative and student functionalities.
- **Sprint 4 (Integration & Testing):** Connecting the UI with the database and performing end-to-end system validation.

Advantages of Agile Methodology

- Faster development through small, focused iterations
- Easy to accommodate changing requirements during development
- Continuous testing at each sprint improves overall quality
- Better collaboration and communication within the team
- Early delivery of working features for feedback and validation

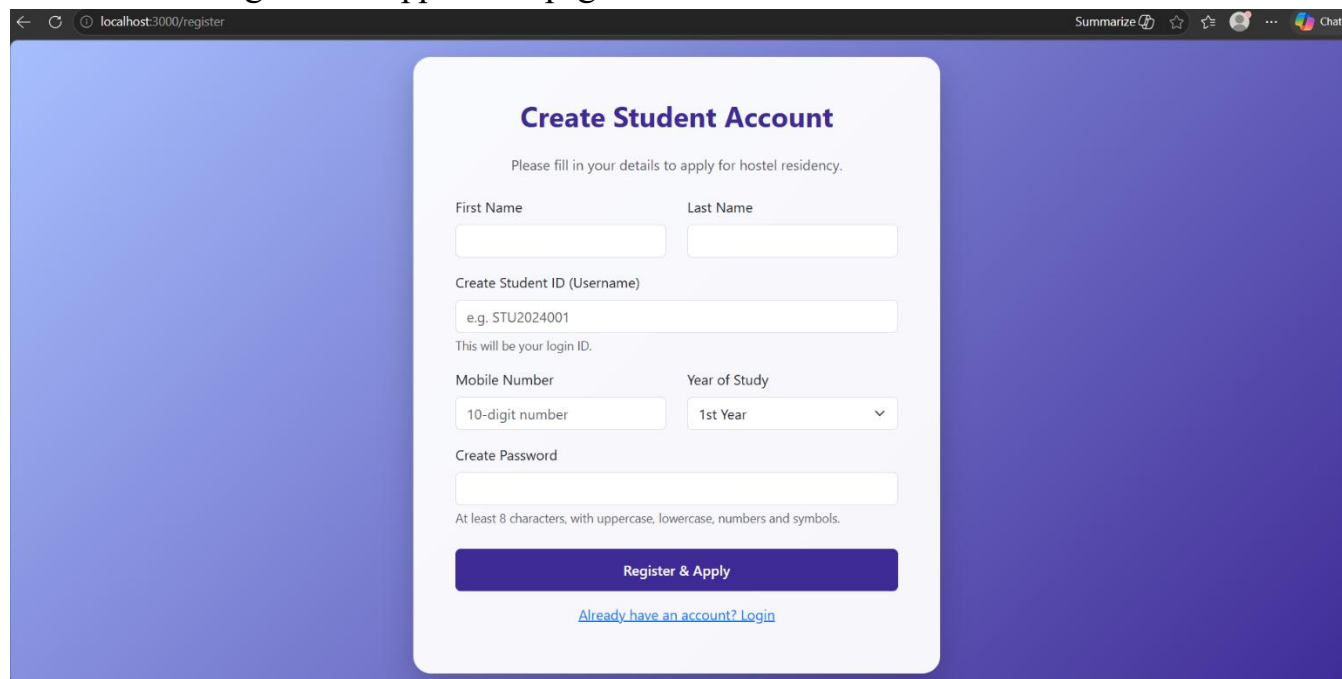
SCREENSHOTS

- Home Page



The screenshot shows the Home Page of the HostelPortal. The browser address bar indicates the URL is localhost:3000. The page has a purple header with the text "HostelPortal" and "Home". On the right side of the header are "Login" and "Register" buttons. The main content area features a large heading "Efficient Hostel Management" followed by a subtext: "Apply for residency, manage your profile, and stay updated with your hostel status through our integrated portal." Below this are three buttons: "Student Registration" (purple), "Student Login" (blue outline), and "Admin Login" (blue). A note below the buttons states: "New students must register and wait for admin approval." To the right is a "Visitor Entry" form with fields for "Visitor Name" (text input), "Whom to Visit (Resident Student)" (dropdown menu), and a green "Register Entry" button. At the bottom, there are three icons: a person with a plus sign, a shield, and a clock.

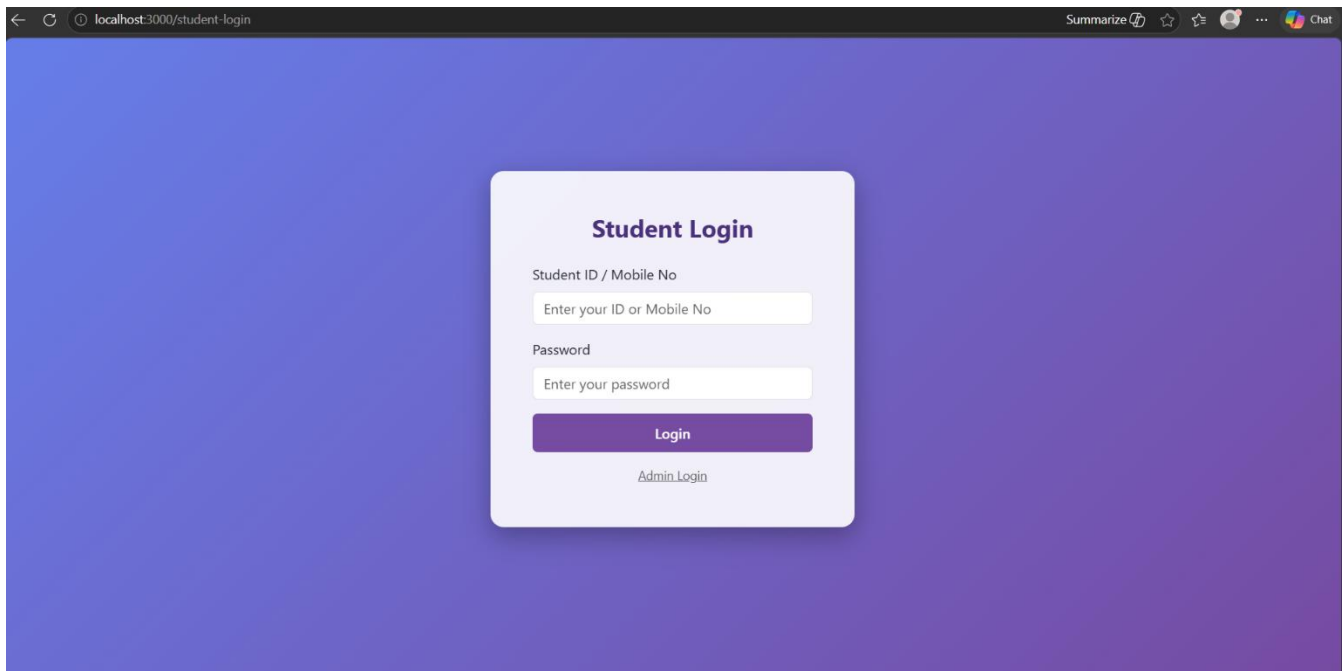
- Student register and application page



The screenshot shows the "Create Student Account" page. The browser address bar indicates the URL is localhost:3000/register. The page has a purple header with "Summarize" and "Chat" buttons. The main content area features a heading "Create Student Account" followed by a subtext: "Please fill in your details to apply for hostel residency." Below this are several form fields: "First Name" and "Last Name" (text inputs), "Create Student ID (Username)" (text input with a hint "e.g. STU2024001" and a note "This will be your login ID."), "Mobile Number" (text input with a hint "10-digit number"), and "Year of Study" (dropdown menu with "1st Year" selected). Below these is a "Create Password" text input with a note "At least 8 characters, with uppercase, lowercase, numbers and symbols." At the bottom is a purple "Register & Apply" button and a link "Already have an account? Login".

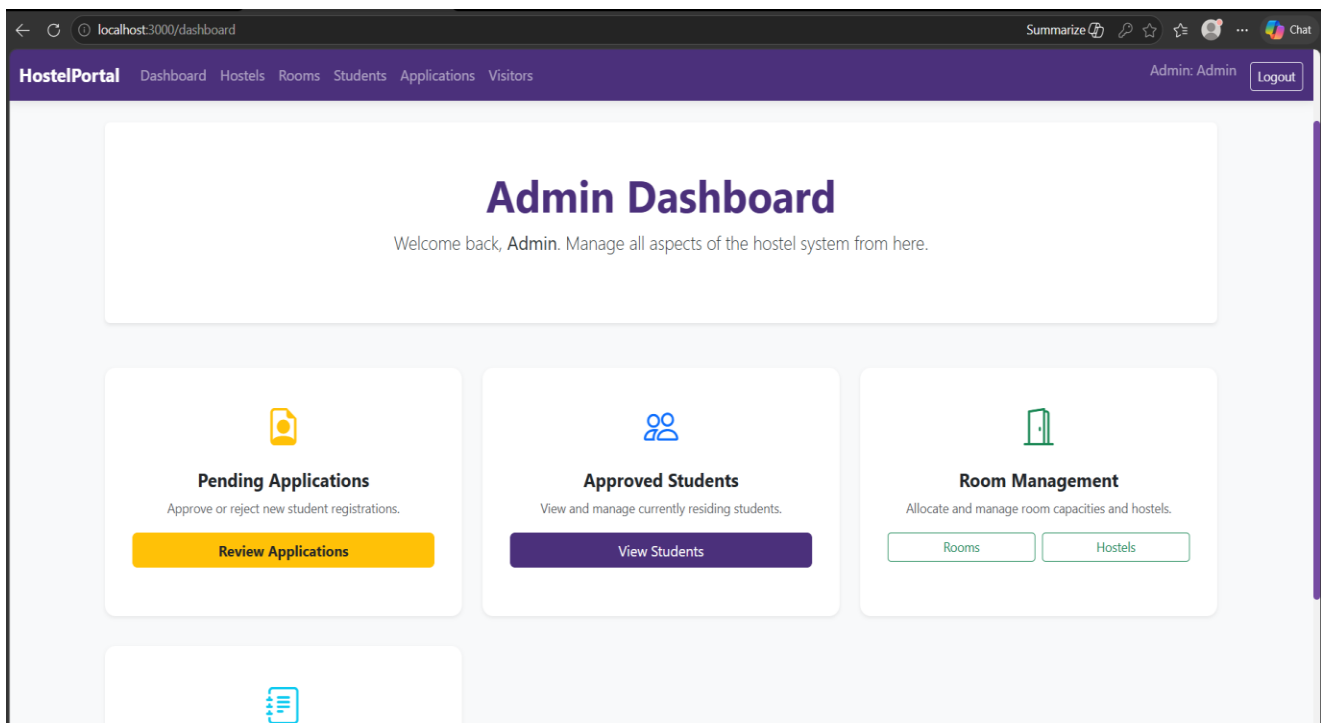
Hostel Room Allocation System

- Student Login Page



A screenshot of a web browser showing the 'Student Login' page. The browser's address bar displays 'localhost:3000/student-login'. The page has a purple gradient background. In the center, there is a white card with the title 'Student Login'. Below the title, there are two input fields: 'Student ID / Mobile No' with the placeholder text 'Enter your ID or Mobile No', and 'Password' with the placeholder text 'Enter your password'. Below these fields is a purple 'Login' button. At the bottom of the card, there is a link labeled 'Admin Login'.

- Admin Dashboard



A screenshot of a web browser showing the 'Admin Dashboard' page. The browser's address bar displays 'localhost:3000/dashboard'. The page has a purple header bar with the text 'HostelPortal' and a navigation menu with links: 'Dashboard', 'Hostels', 'Rooms', 'Students', 'Applications', and 'Visitors'. On the right side of the header, it says 'Admin: Admin' and has a 'Logout' button. The main content area has a white background with the title 'Admin Dashboard' and a welcome message: 'Welcome back, Admin. Manage all aspects of the hostel system from here.' Below this, there are three main sections, each with an icon and a title: 'Pending Applications' (with a document icon), 'Approved Students' (with a group of people icon), and 'Room Management' (with a building icon). Each section has a description and a button: 'Review Applications' (yellow), 'View Students' (purple), and 'Rooms' and 'Hostels' (green). At the bottom, there is a fourth section with a document icon and a list of items.

Hostel Room Allocation System

localhost:3000/students/applications

HostelPortal Dashboard Hostels Rooms Students Applications Visitors Admin: Admin Logout

Pending Registrations

Review and approve new student applications.

2 Pending

STUDENT ID	NAME	MOBILE	YEAR	ACTIONS
S102	Jane Smith	4445556666	Year 3	<button>Approve & Allocate</button> <button>Reject</button>
S103	Peter Jones	7778889999	Year 1	<button>Approve & Allocate</button> <button>Reject</button>

localhost:3000/students

HostelPortal Dashboard Hostels Rooms Students Applications Visitors Admin: Admin Logout

Approved Students

Add New Student

ID	First Name	Last Name	Mobile No.	Year of Study	Room No.	Hostel	Actions
S101	John	Doe	1112223333	2	101	Alpha Hostel	<button>Edit</button> <button>Delete</button>
020	basava	Raj	1234567890	3	102	Alpha Hostel	<button>Edit</button> <button>Delete</button>
058	hemanth ks	ks	9964555666	3	101	Alpha Hostel	<button>Edit</button> <button>Delete</button>
1cd23is036	virat	h s	1234567890	3	102	Alpha Hostel	<button>Edit</button> <button>Delete</button>

TESTING APPROACH

- **Unit Testing:** Validating individual SQL queries to ensure accurate data retrieval (e.g., room counts).
- **Integration Testing:** Verifying the seamless flow of data from frontend forms into the SQLite database.
- **System Testing:** Validating the complete "Application to Allocation" pipeline to ensure logic prevents over-allocation of rooms.

TOOLS USED

- **IDE:** Visual Studio Code (VS Code).
- **Runtime:** Node.js Runtime Environment.
- **Database Tools:** SQLite3 CLI and SQLite Browser for schema management.
- **Version Control:** Git for source code management and version tracking.

CHALLENGES FACED

- **Data Synchronization:** Ensuring room occupancy counts update instantly across the database upon student approval.
- **Complexity of Queries:** Constructing advanced SQL JOINS to aggregate student, hostel, and room details into a single unified view.

CONCLUSION

The **Hostel Room Allocation System** successfully transitions hostel management from a manual, error-prone process to an automated and secure digital platform. By leveraging the performance of **Node.js** and the simplicity of **SQLite**, the system provides a robust tool for administrators while significantly improving the transparency and experience for students. Future enhancements may include online fee payment integration and automated email notification modules.